

Mid - PLSQL Loan Scheduling

```
CREATE TABLE CUSTOMERS
(
  CID NUMBER(10,0),
  NAME VARCHAR2(50 CHAR) NOT NULL,
  CONSTRAINT PK_CUSTOMERS PRIMARY KEY(CID)
);

CREATE TABLE LOANS
(
  LID VARCHAR2(20) PRIMARY KEY,
  NAME VARCHAR2(20),
  PRATE NUMBER(4,2),      -- PROFIT RATE IN PCT PER
ANUM
  INS_SIZE NUMBER(5,0),   -- IN NO OF DAYS, GAP
BETWEEN EACH INSTALLATION
  NOINSTALLMENT NUMBER(2,0), -- NO OF TOTAL
INSTALLMENT
  MAX_AMOUNT NUMBER(15,2) -- MAX AMOUNT
PERMISSIBLE
);

CREATE TABLE LOAN_MASTER
(
  CLID VARCHAR2(20),      --PK
  LID VARCHAR2(20),      --FK
  CID NUMBER(10,0),      --FK
  LOAN_TOTAL_AMOUNT NUMBER(10,2), -- THIS WILL
DETERMINE WHICH TYPE OF LOAN HE/SHE WILL RECEIVE
  DTTIME DATE,
  CONSTRAINT PK_LOANMASTER PRIMARY KEY(CID),
  CONSTRAINT FK_LOANMASTER1 FOREIGN KEY(LID)
REFERENCES LOANS,
  CONSTRAINT FK_LOANMASTER2 FOREIGN KEY(CID)
REFERENCES CUSTOMERS
);

CREATE TABLE LOAN_DETAILS
(
  TID NUMBER(15,6),      --YYMMDD.NNNN, PK
```

```

CLID VARCHAR2(20),          --PK
INS_NO NUMBER(3,0),         -- begins from 1 to upto N
INS_AMOUNT_PRN NUMBER(15,2), -- PRINCIPLE AMOUNT
(fixed amount)
INS_AMOUNT_PRF NUMBER(15,2), -- PROFIT AMOUNT
(extra amount, variable)
INS_DATE_SCH DATE,          --date of payment (scheduled)
CONSTRAINT PK_LOANDETAILS PRIMARY KEY(TID),
CONSTRAINT FK_LOANDETAILS1 FOREIGN KEY(CLID)
REFERENCES LOAN_MASTER
);

```

Your task is to write subprogram in PL/SQL as per the following requirements:

Input: Customer ID and Amount of Loan Applied for.

Output: It will schedule the loan accordingly (i.e. 2 loan tables will be populated).

Algorithm: The amount will determine the type of loan the customer should get. Then it will make a standard schedule (depending on the type, as per the amount paid for). The payment will be as such, where the INS_AMOUNT_PRN is fixed in each round while INS_AMOUNT_PRF will be initialized to 0. Date will be determined as per loan properties (i.e. INS_SIZE value in no. of days).

Solve this with very clear explanation of everything step by step

Absolutely! Let's break this down step-by-step.

You need a PL/SQL **subprogram** (let's use a procedure) that will:

- **Take as input:** Customer ID (`p_cid`) and Loan Amount (`p_amt`)
- **Figure out:** Which loan product this customer qualifies for (based on amount)
- **Create entries:** In `LOAN_MASTER` (loan issued), and then in `LOAN_DETAILS` (the full payment schedule—principal per installment, profit as zero, scheduled dates)

- **Use the properties** (like installment count, gap between installments) of the selected loan product

Let's go step by step, explaining as we go.

Step 1: Identify the Loan Product

We need to pick the best loan type whose **MAX_AMOUNT** is greater than or equal to the applied amount.

We'll pick the smallest such loan product (if more than one qualify), for fairness.

SQL:

```
sql

SELECT *
INTO v_loan
FROM (
  SELECT *
  FROM LOANS
  WHERE MAX_AMOUNT >= p_amt
  ORDER BY MAX_AMOUNT
) WHERE ROWNUM = 1;
```

We'll fetch LID, PRATE, INS_SIZE, NOINSTALLMENT, MAX_AMOUNT.

Step 2: Create a new LOAN_MASTER record

- Create a unique **CLID** (could be a sequence or generated key—here, use `CID || '_' || TO_CHAR(SYSDATE, 'YYYYMMDDHH24MISS')`)
- Set **LID** (from above), **CID** , **LOAN_TOTAL_AMOUNT** (p_amt), **DTIME** (SYSDATE)

SQL:

```
sql

INSERT INTO LOAN_MASTER (CLID, LID, CID, LOAN_TOTAL_AMOUNT, DTIME)
VALUES (v_clid, v_lid, p_cid, p_amt, SYSDATE);
```

Step 3: Calculate Installment Amounts and Dates

- **Principal per installment:**

`PRINCIPAL = p_amt / NOINSTALLMENT` (rounded as needed)

- **Profit per installment:**

`PROFIT = 0` (as per your rules)

- **Installment dates:**

1st date = today

Each next = previous + INS_SIZE days

Step 4: Populate LOAN_DETAILS table

For each installment from 1 to N:

- TID: YYMMDDNNNN (can use sequence or construct—here, simulate with

`TO_CHAR(SYSDATE, 'YYMMDD') || LPAD(i, 4, '0')`

- CLID: As above
- INS_NO: 1, 2, ..., N
- INS_AMOUNT_PRN: Calculated above
- INS_AMOUNT_PRF: 0
- INS_DATE_SCH: Calculated date

SQL in a loop:

sql

```
FOR i IN 1..v_no_installments LOOP
    v_ins_date := v_start_date + (i-1)*v_ins_size;
    INSERT INTO LOAN_DETAILS (TID, CLID, INS_NO, INS_AMOUNT_PRN, INS_AMOUNT_PRF,
INS_DATE_SCH)
    VALUES (...);
END LOOP;
```

Complete PL/SQL Procedure

Here's the full annotated code:

sql

```
CREATE OR REPLACE PROCEDURE schedule_loan (
  p_cid IN CUSTOMERS.CID%TYPE,
  p_amt IN NUMBER
)
IS
  -- Variables for selected loan product
  v_lid      LOANS.LID%TYPE;
  v_prate    LOANS.PRATE%TYPE;
  v_ins_size LOANS.INS_SIZE%TYPE;
  v_no_inst  LOANS.NOINSTALLMENT%TYPE;
  v_max_amount LOANS.MAX_AMOUNT%TYPE;

  -- Variables for new master record
  v_clid     VARCHAR2(20);
  v_start_date DATE := SYSDATE;

  -- For loop
  v_principal NUMBER;
  v_profit     NUMBER := 0;
  v_ins_date   DATE;
  v_tid        NUMBER(15,6);
BEGIN
  -- 1. Find suitable loan product
  SELECT LID, PRATE, INS_SIZE, NOINSTALLMENT, MAX_AMOUNT
  INTO v_lid, v_prate, v_ins_size, v_no_inst, v_max_amount
  FROM (
    SELECT *
    FROM LOANS
    WHERE MAX_AMOUNT >= p_amt
    ORDER BY MAX_AMOUNT
  )
  WHERE ROWNUM = 1;

  -- 2. Generate CLID (you can adjust this logic for uniqueness)
  v_clid := TO_CHAR(p_cid) || '-' || TO_CHAR(SYSDATE, 'YYYYMMDDHH24MISS');

  -- 3. Insert into LOAN_MASTER
```

```

INSERT INTO LOAN_MASTER (CLID, LID, CID, LOAN_TOTAL_AMOUNT, DTTIME)
VALUES (v_clid, v_lid, p_cid, p_amt, SYSDATE);

-- 4. Calculate installment principal amount (fixed)
v_principal := ROUND(p_amt / v_no_inst, 2);

-- 5. Loop to insert LOAN_DETAILS
FOR i IN 1..v_no_inst LOOP
    v_ins_date := v_start_date + (i-1)*v_ins_size;
    -- Construct TID as YYMMDDNNNN (not perfectly unique in production)
    v_tid := TO_NUMBER(TO_CHAR(SYSDATE, 'YYMMDD') || LPAD(i, 4, '0'));
    INSERT INTO LOAN_DETAILS (
        TID, CLID, INS_NO, INS_AMOUNT_PRN, INS_AMOUNT_PRF, INS_DATE_SCH
    )
    VALUES (
        v_tid, v_clid, i, v_principal, v_profit, v_ins_date
    );
END LOOP;
COMMIT;
EXCEPTION
    WHEN NO_DATA_FOUND THEN
        DBMS_OUTPUT.PUT_LINE('No suitable loan product found for the amount.');
```

```

    WHEN OTHERS THEN
        DBMS_OUTPUT.PUT_LINE('Error: ' || SQLERRM);
    ROLLBACK;
END;
/
```

Explanation Recap

1. Pick the right loan product (by amount).
2. Create a unique master loan record for the customer.
3. Calculate the fixed principal installment and set profit to zero.
4. Schedule installments: for each, create a detail row with correct date.
5. Commit everything, and handle exceptions gracefully.

How to Call This Procedure

```
sql
```

```
BEGIN
```

```
    schedule_loan(101, 50000); -- Example: Customer ID 101 applies for 50,000
```

```
END;
```

Let me know if you want real code for your DB (with sequences, more robust IDs, etc.), or if you want to see this as a function, or want to add custom logic for profit calculations!

now show me exactly how it works in step by step details with examples

Absolutely! Let's walk through exactly **how the procedure works step by step**, using a realistic example with sample data. This will make everything crystal clear.

Step 0: Example Setup

Suppose your database tables contain:

Sample LOANS Table

LID	NAME	PRATE	INS_SIZE	NOINSTALLMENT	MAX_AMOUNT
LN1	Micro	10	30	6	20000.00
LN2	Small	9	30	12	100000.00
LN3	Medium	8	30	24	500000.00

Sample CUSTOMERS Table

CID	NAME
101	Asif Ahmed
102	Rita Karim

Now, Suppose

Customer Rita Karim (CID=102) applies for a loan of 50,000.

Step 1: Call the Procedure

sql

```
BEGIN
  schedule_loan(102, 50000);
END;
```

Step 2: Find Suitable Loan Product

- **Requested Amount:** 50,000

We search the `LOANS` table for a product whose `MAX_AMOUNT` $\geq$ 50,000.

- LN1 (Micro): 20,000 (not enough)
- LN2 (Small): 100,000 (**enough**)
- LN3 (Medium): 500,000 (**also enough**)

We pick the loan with the smallest `MAX_AMOUNT` that still qualifies (LN2).

Selected Loan:

- LID: LN2

- PRATE: 9
 - INS_SIZE: 30
 - NOINSTALLMENT: 12
 - MAX_AMOUNT: 100,000
-

Step 3: Insert into LOAN_MASTER

- Generate unique **CLID**: e.g., '102_20250702180001' (CID + date/time string)
- **LID**: 'LN2'
- **CID**: 102
- **LOAN_TOTAL_AMOUNT**: 50000
- **DTIME**: current date/time (say, 2025-07-02)

New LOAN_MASTER Row:

CLID	LID	CID	LOAN_TOTAL_AMOUNT	DTIME
102_20250702180001	LN2	102	50000.00	2025-07-02

Step 4: Calculate Installment Amounts and Dates

- **Installment Count**: 12
- **Principal per Installment**: $50000 / 12 = 4166.67$ (rounded to two decimals)
- **Profit per Installment**: 0 (as per your requirement)
- **Installment Interval**: 30 days

Installment No.	Date	Principal	Profit
1	2025-07-02	4166.67	0.00
2	2025-08-01	4166.67	0.00
3	2025-08-31	4166.67	0.00
...	...	...	...
12	2026-06-29	4166.67	0.00

(Each next installment date is previous date + 30 days)

Step 5: Insert into LOAN_DETAILS (For Each Installment)

For each installment (1 to 12):

- **TID:** For example, for Installment 1, TID = 2507020001 (YYMMDDNNNN)
(You may use sequence/logic as needed for real uniqueness.)
- **CLID:** '102_20250702180001'
- **INS_NO:** 1, 2, ..., 12
- **INS_AMOUNT_PRN:** 4166.67
- **INS_AMOUNT_PRF:** 0
- **INS_DATE_SCH:** As above

First Three Rows Example:

TID	CLID	INS_NO	INS_AMOUNT_PRN	INS_AMOUNT_PRF	IN
2507020001	102_20250702180001	1	4166.67	0.00	20
2507020002	102_20250702180001	2	4166.67	0.00	20
2507020003	102_20250702180001	3	4166.67	0.00	20

TID	CLID	INS_NO	INS_AMOUNT_PRN	INS_AMOUNT_PRF	IN
...	...	...	...	...	...
2507020012	102_20250702180001	12	4166.67	0.00	20

Step 6: Commit the Transaction

All rows are inserted into **LOAN_MASTER** and **LOAN_DETAILS**.

Summary Table of What Happened

1. Procedure is called with (102, 50000)
2. System chooses loan LN2 (Small) because $\text{MAX_AMOUNT } 100,000 \geq 50,000$
3. System creates a master record for this customer/loan
4. System creates 12 installments, each with fixed principal (4166.67), profit (0), and schedule (every 30 days)
5. System populates **LOAN_DETAILS** with all these scheduled payments

What if Customer Applies for Different Amounts?

- If customer applied for **10,000** → would match LN1 (Micro), 6 installments, 1666.67 each, every 30 days.
- If customer applied for **300,000** → would match LN3 (Medium), 24 installments, 12,500 each, every 30 days.

Visualization

- **You input:** Customer ID and loan amount
 - **System picks:** The best loan for that amount (smallest MAX_AMOUNT $\geq$ your amount)
 - **It creates:** One LOAN_MASTER record and N LOAN_DETAILS records (installments)
 - **Each installment:** Has same principal, 0 profit, scheduled date based on INS_SIZE
-

If you want to try with your own data, just call the procedure with any valid CID and amount. You'll see the tables fill up automatically, as shown above.

Let me know if you want to see the actual PL/SQL outputs, or if you want a script that queries the results after execution!